

Ioannis P. Rizos

publishing as *J. Rizos*

Professor of Theoretical Physics

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Professor of Theoretical Physics at the University of Ioannina, on the faculty since 1996 and full professor since 2014. Author of **71 papers in international peer-reviewed journals** (5,173 citations, *h*-index 36); Marie Curie Fellow (1995–96, 2003) and Scientific Associate at CERN (2013). Nearly thirty years of teaching, examining and curriculum design in theoretical physics — quantum mechanics, classical physics, electrodynamics — and in computational methods, including five years on the Hellenic Open University’s M.Sc. in Data Science and Machine Learning, taught in English. Former Head of the Department of Physics (2013–2017) and President of the Southeastern European Network in Mathematical and Theoretical Physics (SEENET-MTP) since 2018.

Research Interests

String phenomenology and the construction of realistic four-dimensional string vacua; classification of heterotic superstring models in the free-fermionic and orbifold formulations; non-supersymmetric string model building and the decompactification problem; grand unified theories, neutrino masses and proton stability; string effective actions, higher-curvature gravity and non-singular cosmological solutions. Computational and machine-learning approaches — genetic algorithms, quantum annealing and classification techniques — applied to the exploration of the string landscape.

Contact

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Education

Nov 1991 **Ph.D. in Physics**, University of Ioannina. Thesis: *Construction and Phenomenological Analysis of Supersymmetric Unified Models from Four-Dimensional Superstring Theories*. Advisor: Prof. K. Tamvakis.

1986 – 1991 Doctoral researcher (Special Postgraduate Scholar, E.M.Y.), Department of Physics, University of Ioannina.

1984 – 1985	Graduate studies in Physics, University of Crete.
1980 – 1984	B.Sc. in Physics, Department of Physics, University of Ioannina.

| Academic Appointments

2014 – present	Professor , Department of Physics, University of Ioannina (elected June 2014; appointed 4 September 2014).
2004 – 2014	Associate Professor , Department of Physics, University of Ioannina (elected November 2004; appointed June 2005).
1996 – 2004	Assistant Professor , Department of Physics, University of Ioannina — elected February 1994 while a postdoctoral researcher at SISSA/ISAS; took up duties October 1996 following military service and a further research appointment at SISSA; tenured April 2000.
1993 – 1994	Postdoctoral Researcher, International School for Advanced Studies (SISSA/ISAS), Trieste, Italy.
1991 – 1993	Postdoctoral Researcher, Centre de Physique Théorique, École Polytechnique, Palaiseau, France.

| Visiting Positions

Jun – Sep 2013	Scientific Associate , CERN, Geneva, Switzerland.
Feb – Jul 2003	Visiting Professor (<i>Professeur Invité</i>), Centre de Physique Théorique, École Normale Supérieure, Paris, France (<i>Marie Curie Fellow</i>).
Oct 2000 – Jul 2001	Visiting Professor (<i>Professeur Invité</i>), Centre de Physique Théorique, École Polytechnique, Palaiseau, France.
May – Aug 2000	<i>Corresponding Associate</i> , CERN, Geneva, Switzerland.
Nov 1995 – Oct 1996	Visitor, SISSA/ISAS, Trieste, Italy (<i>Marie Curie Fellow</i>).
Oct – Nov 1995	Visitor, CERN, Geneva, Switzerland.

| Fellowships and Awards

2003	Marie Curie Fellow, European Union (6 months).
1995 – 1996	Marie Curie Fellow, European Union (6 months).
1991 – 1993	Fellow of the European Community (<i>Science Programme</i>).
1986 – 1991	Special Postgraduate Scholar (E.M.Y.), Department of Physics, University of Ioannina.
1981 – 1984	Fellow of the Greek State Scholarships Foundation (I.K.Y.), three consecutive years of undergraduate study.

| Research Funding and Projects

High Energy Physics

2019 – 2023	COST Action CA18108 <i>QG-MM: Quantum Gravity Phenomenology in the Multi-Messenger Approach</i> .
2012 – 2016	<i>ARISTEIA</i> : Particle Physics and Cosmology in the LHC Era, co-funded by the European Social Fund.
2012 – 2016	<i>THALES</i> : Theoretical Elementary Particle Physics and Cosmology in the Light of the LHC, co-funded by the ESF.
2012 – 2015	Particle Phenomenology and Cosmology in the LHC Era (K.E. 80906), NSRF 2007–13.
2009 – 2013	<i>ITN Unification in the LHC Era</i> , PITN-GA-2009-237920, European Union.
2006 – 2009	<i>RTN UniverseNet</i> , MRTN-CT-2006-035863, European Union.
2005 – 2007	<i>HERAKLEITOS</i> : Embedding of Unified Gauge Theories in Higher-Dimensional Spaces, EPEAEK.
2004 – 2009	<i>RTN The Quest for Unification: Theory Confronts Experiment</i> , MRTN-CT-2004-503369, European Union.
2001 – 2004	<i>RTN Supersymmetry and the Early Universe</i> , HPRN-CT-2000-00152, European Union.
2000 – 2004	<i>RTN network Across the Present Energy Frontier: Probing the Origin of Mass</i> , HPRN-CT-2000-00148, European Union — member of the research team and scientific coordinator and representative of the Ioannina node .
1997 – 1999	<i>TMR Beyond the Standard Model</i> , FMRX-CT96-0090, European Union.
1991 – 1993	<i>Science Programme</i> , European Union (fellow).

Information Technology, Networks and e-Learning — as Scientific Coordinator

Scientific Coordinator renders the Greek role Επιστημονικός Υπεύθυνος: the academic bearing scientific and administrative responsibility for a project before the university's Research Committee. These are ICT infrastructure and e-learning projects rather than research grants.

2023 – 2026	Scientific Coordinator , <i>Extension and Development of the Greek School Network</i> (K.E. 83427), Ministry of Education.
2016 – present	Scientific Coordinator , <i>Acquisition of Academic Teaching Experience by Young Postdoctoral Scholars at the University of Ioannina</i> , OP Human Resources Development, Education and Lifelong Learning.
2012 – 2015	Scientific Coordinator , <i>Open Academic Courses of the University of Ioannina</i> , NSRF 2007–13.
2011 – 2015	Scientific Coordinator , <i>STIRIZO — Horizontal project supporting schools, teachers and pupils on the road to the digital school</i> , NSRF 2007–13.
2006 – 2009	Scientific Coordinator , <i>Advanced Telematic Services for Education and Research at the University of Ioannina</i> , OP Information Society.
2005 – 2008	Scientific Coordinator , <i>Epirus Web Portal</i> (K.E. 11967 design and implementation; K.E. 11968 equipment), Ministry of Education.
2004 – 2009	Scientific Coordinator , <i>Technical Support for the Construction of Metropolitan Broadband Fibre-Optic Networks in the Region of Epirus</i> , OP Information Society.
2004 – 2006	Scientific Coordinator , <i>Advanced Telematic Services for Secondary Education</i> (prefectures of Ioannina, Arta, Preveza, Thesprotia, Corfu and Lefkada), OP Information Society.

Information Technology and e-Learning — as participant

2001 – 2005	Advanced Telematic Services for Primary Education; School Information Systems Help Desk; <i>DIKTYOTHEITE</i> (EU, Information Society); equipment and upgrading of tele-teaching and computer laboratories — OP Information Society (various calls).
2001 – 2003	Advanced Telematic Services at the University of Ioannina; <i>EDUNET</i> Panhellenic Network for Education (Ioannina region); wireless connectivity for school units.
2003 – 2005	Equipment and Networking in Primary Education, OP Information Society.

2005 – 2008	New Postgraduate Programmes of the University of Ioannina — <i>New Technologies and Research in Physics Education</i> , Ministry of Education.
2007 – 2008	Training of Trainers Programmes (PAKE), Ministry of Education — teaching of General and Distance-Education methodology modules for the PE04 discipline (64 teaching hours).
2019	Teacher Training for the Exploitation and Application of Digital Technologies, Ministry of Education.
1999 – 2000	Distance Support for Mathematics Teaching through Networks and Computing Tools, Ministry of Education.

Teaching Experience

University of Ioannina — Undergraduate courses, Department of Physics

2018 – present	Introduction to Quantum Field Theory (elective; 7 semesters).
2015 – present	Quantum Theory II (compulsory; 10 semesters).
2011 – 2017	Classical Electrodynamics I (compulsory; 5 semesters).
2004 – 2025	Classical Mechanics II (compulsory; 12 semesters).
2004 – 2011	Classical Mechanics I (compulsory; 7 semesters).
2002 – present	<i>Mathematics and Physics with Computers</i> (elective; designed and taught; 22 semesters).
2002 – 2011	Mathematics for Physicists (elective; 7 semesters).
2002 – 2011	<i>New Technologies in Education</i> (elective; designed and taught; 8 semesters).
2003 – 2005	<i>Internet Applications</i> (elective; designed and taught; 2 semesters).
1997 – 2011	Group Theory (7 semesters).
1997 – 2002	Computer Applications in Physics Research and Teaching (elective; 6 semesters).

University of Ioannina — Graduate courses

2014 – 2018	Electronic Learning Environments and Applications in Physics Teaching, M.Sc. <i>New Technologies and Research in Physics Education</i> (4 semesters).
2014 – 2016	Educational Technologies and Applications in Physics Teaching, same programme (2 semesters).
2005 – 2014	Modern Technologies in the Service of Education (9 semesters) and Development of Distance-Learning Methods (9 semesters), same programme.
2004 – 2007	Classical Mechanics, M.Sc. in Meteorology, Climatology and Atmospheric Environment Physics (2 semesters).
2002 – 2005	Computational Methods in Physics (elective), M.Sc. in Physics (2 semesters).

Hellenic Open University — M.Sc. in Data Science and Machine Learning (taught in English)

2021 – 2026	DAMA50 — Mathematics for Machine Learning. Tutor (S.E.P.), five academic years; Module Coordinator 2022–23 and 2023–24.
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Hellenic Open University — B.Sc. Programme in Natural Sciences

2017 – 2021	FYE40 — Quantum Physics. Tutor, four academic years; Module Coordinator 2017–18 through 2021–22.
2004 – 2017	FYE34 — Classical Physics II. Tutor, thirteen academic years; Module Coordinator 2015–16 and 2016–17.

Development of Educational Material for Distance Learning

- 24-hour podcast series for PSF60 (Quantum Physics), Hellenic Open University (2008).
- Production of WEBCAST material for PSF60 (Quantum Physics), Hellenic Open University (2007–08).
- Critical review and evaluation of WEBCAST material for PSF50 (Classical Physics), Hellenic Open University (2007–08).
- Development of multiple-choice question banks for FYE34 (Classical Physics II), Hellenic Open University (2007–08).
- Complete digital open courses *Classical Electrodynamics I* and *Mathematics and Physics with Computers* for the institutional Open Academic Courses platform (opencourses.gr), University of Ioannina (2012–15).

Research Supervision

2004 – present	Supervision of 15 M.Sc. theses within the graduate programmes of the Department of Physics, University of Ioannina, and of numerous undergraduate diploma theses. Supervision of one doctoral dissertation ; member of three-member advisory committees for M.Sc. theses and doctoral dissertations, and of seven-member doctoral examination committees.
1999 – 2009	Supervision of five diploma theses in the interdisciplinary programme <i>Sciences and Culture</i> , University of Ioannina.

Academic Leadership and Administration

Department and University of Ioannina

2022 – present	Chair , Internal Evaluation Group (OMEA), Department of Physics.
2013 – 2017	Head , Department of Physics.
2013 – 2017	Member of the Deanery, School of Sciences.
2013 – 2014	Member of the Senate of the University of Ioannina.
2012 – 2020	Director , M.Sc. programme <i>New Technologies and Research in Physics Education</i> — Chair of the programme's Coordinating Committee (Συντονιστική Επιτροπή Μεταπτυχιακών Σπουδών), responsible for formulating and proposing the programme's curriculum, admissions, teaching assignments and quality procedures for approval by the Departmental Assembly.
2012 – 2017	Chair , Undergraduate Curriculum Committee, Department of Physics.
2009 – 2013	Deputy Head , Department of Physics.
2007 – 2013	Director , Division of Theoretical Physics.
2006 – 2011	Member of the Research Committee of the University of Ioannina, representing the Department of Physics.
2002 – 2013	Member, Committee for the Evaluation of Teaching, Department of Physics.
1998 – 2013	Member, Study Guide Committee, Department of Physics; responsible for the online edition.

Faculty appointment and promotion committees

ongoing Member of **electoral bodies** (*εκλεκτορικά σώματα*) for academic appointments and promotions, and of the three-member **evaluation committees** (*εισηγητικές επιτροπές*) responsible for assessing candidates' research records and drafting the written evaluation report on which the decision is based. Three evaluation committees in 2025–26: the Department of Physics, University of Ioannina; the Department of Physics, Aristotle University of Thessaloniki; and the National Centre for Scientific Research *Demokritos*.

Information technology

2005 – 2015 **Scientific Director**, Network Operations Centre, University of Ioannina.
2005 – 2013 **Chair**, Computer Committee, Department of Physics (member 2000–2005).
2001 – 2011 Member, Scientific Committee of the Network Operations Centre, University of Ioannina.
1997 – 2017 Responsible for the website of the Department of Physics; system administrator and webmaster of the Division of Theoretical Physics (1997–2013).

Hellenic Open University

2021 – 2024 Member, Curriculum Committee (E.P.S.) of the M.Sc. programme *Data Science and Machine Learning*.

Scientific Networks and Conference Organisation

2018 – present **President**, Southeastern European Network in Mathematical and Theoretical Physics (SEENET-MTP), seenet-mtp.info. Re-elected July 2025 for a further two-year term.
2017 – present Member, Scientific Advisory Committee, SEENET-MTP.

Conference and school organisation

2019 **Main organiser**, BS2019: SEENET-MTP Balkan School on High Energy and Particle Physics, Ioannina, 3–10 June 2019.
2017 **Main organiser**, HEP 2017: Recent Developments in High Energy Physics and Cosmology, Ioannina, 6–9 April 2017.
2016 **Main organiser**, STRINGPHENO 2016: 15th Conference in the String Phenomenology Series, Ioannina, 20–24 June 2016.
2015 Organising Committee, PLANCK 2015: 18th International Conference from the Planck Scale to the Electroweak Scale, Ioannina.
2015 **Main organiser**, *Beyond the Standard Models of Particle Physics and Cosmology*, workshop in honour of K. Tamvakis, Ioannina, 24 May 2015.
2012, 2006 Organising Committee, HEP2012 and HEP2006: Recent Developments in High Energy Physics and Cosmology, Ioannina.
2005 Organising Committee, CORFU2005: Corfu Summer Institute on Elementary Particle Physics, and *The Quest for Unification: Theory Confronts Experiment*, Corfu.
2001 Organising Committee, South European School on Elementary Particle Physics, Corfu (EC 5th Framework, HPCF-2001-00052-01), and the 4th RTN meeting *Across the Present Energy Frontier*.
2000 Organising Committee, Annual Workshop of the Hellenic Society for the Study of High Energy Physics, Ioannina.

Editorial and Refereeing Activity

2021 – present	Editorial Board, <i>Particles</i> .
2020 – present	Editorial Board, <i>International Journal of Modern Physics A</i> .
2020 – present	Editorial Board, <i>Modern Physics Letters A</i> .
2008 – present	Referee, on invitation, for <i>Nuclear Physics B</i> , <i>Journal of High Energy Physics</i> , <i>International Journal of Modern Physics A</i> , <i>Modern Physics Letters A</i> and <i>Universe</i> .

Professional Memberships

1999 – present	Marie Curie Fellows Association.
1990 – present	Hellenic Society for the Study of High Energy Physics.
1986 – present	Union of Greek Physicists.

Languages and Computing Skills

<i>Languages:</i>	Greek (native), English (fluent), French (very good).
<i>Programming:</i>	Python, Mathematica, FORTRAN 77/90/95.
<i>Web & scripting:</i>	HTML, PHP, PERL, Java, JavaScript.
<i>Systems:</i>	UNIX/Linux, Windows, macOS.
<i>Tools:</i>	$\text{\LaTeX}$, Word, Excel, PowerPoint, CorelDraw.

Active use of Python and Mathematica in both research and teaching. Application of optimisation algorithms — genetic algorithms and quantum annealing — to the classification of theoretical models (see *Selected publications on computational methods and machine learning* below).

Research Impact

<i>INSPIRE-HEP:</i>	5,173 citations • <i>h</i> -index 36 https://inspirehep.net/authors/991609?ui-citation-summary=true
<i>Google Scholar:</i>	5,886 citations • <i>h</i> -index 39 https://scholar.google.com/citations?user=7TJ7ZnAAAAAJ
<i>ORCID:</i>	https://orcid.org/0000-0001-7586-1700
<i>Summary:</i>	71 articles in international peer-reviewed journals; 1 invited handbook chapter; 14 papers in international conference proceedings; 5 papers in science-education proceedings; translation and scientific editing of a two-volume university physics textbook (3rd and 4th Greek editions).

Selected Invited Talks and Seminars

Jul 2025	<i>On the Quest for the Standard Model Among String Theory Vacua</i> , BWAM25: SEENET-MTP Workshop and Assessment Meeting, Bucharest, Romania.
Dec 2023	<i>On Non-Supersymmetric String Models</i> , Xmas Theoretical Physics Workshop, Athens.

Sep 2023	<i>On Three-Generation Super No-Scale Models in Heterotic String Theory</i> , CORFU2023, Corfu.
Aug 2023	<i>String Model Building Using Quantum Annealers</i> , BW2023 SEENET-MTP Meeting, Vrnjačka Banja, Serbia.
Aug 2023	<i>On String Model Building in the Free-Fermionic Formulation</i> , Department of Physics, University of Niš, Serbia.
Sep 2022	<i>On 3-Generation Non-Supersymmetric Heterotic String Models</i> , CORFU2022, Corfu.
Sep 2021	<i>Non-Supersymmetric String Phenomenology</i> , SEENET-MTP Workshop BW2021, Belgrade, Serbia.
Jul 2021	<i>On Non-Supersymmetric String Models and the Decompactification Problem</i> , Crete Center for Theoretical Physics, University of Crete.
Jun 2021	<i>On Non-Supersymmetric String Phenomenology</i> , Nuclear and Particle Physics Seminar, University of Athens.
Oct 2019	<i>On Non-Supersymmetric String Phenomenology</i> , Assessment Meeting of the SEENET-MTP-ICTP Project NT-03, ICTP, Trieste, Italy.
Sep 2019	<i>On Non-Supersymmetric String Model Building</i> , CORFU2019, Corfu.
Mar 2018	<i>Aspects of Non-Supersymmetric String Phenomenology</i> , HEP 2018, Athens.
Mar 2017	<i>Aspects of Non-Supersymmetric String Phenomenology</i> , Department of Physics, University of Cyprus, Nicosia.
Jul 2014	<i>Top Quark Mass Coupling and the Classification of $SO(10)$ Heterotic Superstring Vacua</i> , String Phenomenology 2014, Trieste, Italy.
Apr 2013	<i>On the Classification of Heterotic Superstring Models</i> , Division of Theoretical Physics, University of Liverpool, UK.
Jul 2010	<i>Towards Classification of $Z_2 \times Z_2$ Heterotic Superstring Vacua</i> , String Phenomenology 2010, Paris.
Jan 2007	<i>Classification of $Z_2 \times Z_2$ Heterotic Orbifold Models and Spinor-Vector Duality</i> , University of Liverpool, UK.
Jan 2006	<i>The Standard Model and its Extensions from Strings</i> , Centre de Physique Théorique, École Polytechnique, Palaiseau, France.
Dec 2001	<i>A Pati-Salam Scenario from Branes</i> , Inaugural Meeting of the RTN Network Across the Present Energy Frontier, Oxford, UK.
Dec 1993	<i>Construction and Phenomenological Analysis of Superstring Unified Models in the Free-Fermionic Formulation</i> , SISSA/ISAS, Trieste, Italy.

Selected Publications on Computational Methods and Machine Learning

A subset of the works listed below, highlighted for relevance; these items are not additional to the counts given above.

1. **“String Model Building on Quantum Annealers,”** S. A. Abel, L. A. Nutricati and J. Rizos, *Fortsch. Phys.* **71** (2023) 2300167. — Application of quantum annealing to large-scale optimisation problems.
2. **“Genetic Algorithms and the Search for Viable String Vacua,”** S. Abel and J. Rizos, *JHEP* **1408** (2014) 010. — Application of evolutionary algorithms in high-dimensional parameter spaces.
3. **“Towards Machine Learning in the Classification of $Z_2 \times Z_2$ Orbifold Compactifications,”** A. E. Faraggi, G. Harries, B. Percival and J. Rizos, *J. Phys.: Conf. Ser.* **1586** (2020) 012032. — Among the earliest applications of machine-learning techniques to the classification of string models.
4. **“MERLIN: A Portable System for Multidimensional Minimization,”** G. A. Evangelakis, J. Rizos, I. E. Lagaris and I. N. Demetropoulos, *Computer Physics Communications* **46** (1987) 401–415. — Development of a multidimensional optimisation system; a foundational numerical-analysis tool directly relevant to model-training methods.

1. **“The Flipped $SU(5) \times U(1)$ Model from Four-Dimensional Strings,”** I. Antoniadis and J. Rizos, *Universe* **11** (2025) no.2, 44 doi:10.3390/universe11020044
2. **“From free-fermionic constructions to orbifolds and back,”** I. Florakis and J. Rizos, *JHEP* **01** (2024), 151 doi:10.1007/JHEP01(2024)151 [[arXiv:2310.10438](#) [hep-th]].
3. **“String Model Building on Quantum Annealers,”** S. A. Abel, L. A. Nutricati and J. Rizos, *Fortsch. Phys.* **71** (2023) no.12, 2300167 doi:10.1002/prop.202300167 [[arXiv:2306.16801](#) [hep-th]].
4. **“Three-generation super no-scale models in heterotic superstrings,”** I. Florakis, J. Rizos and K. Violaris-Gountonis, *Phys. Lett. B* **833** (2022), 137311 doi:10.1016/j.physletb.2022.137311 [[arXiv:2206.09732](#) [hep-th]].
5. **“Particle physics and cosmology of the string derived no-scale flipped $SU(5)$,”** I. Antoniadis, D. V. Nanopoulos and J. Rizos, *Eur. Phys. J. C* **82** (2022) no.4, 377 doi:10.1140/epjc/s10052-022-10353-6 [[arXiv:2112.01211](#) [hep-th]].
6. **“Super no-scale models with Pati-Salam gauge group,”** I. Florakis, J. Rizos and K. Violaris-Gountonis, *Nucl. Phys. B* **976** (2022), 115689 doi:10.1016/j.nuclphysb.2022.115689 [[arXiv:2110.06752](#) [hep-th]].
7. **“Cosmology of the string derived flipped $SU(5)$,”** I. Antoniadis, D. V. Nanopoulos and J. Rizos, *JCAP* **03** (2021), 017 doi:10.1088/1475-7516/2021/03/017 [[arXiv:2011.09396](#) [hep-th]].
8. **“Doublet-Triplet Splitting in Fertile Left-Right Symmetric Heterotic String Vacua,”** A. E. Faraggi, G. Harries, B. Percival and J. Rizos, *Nucl. Phys. B* **953** (2020), 114969 doi:10.1016/j.nuclphysb.2020.114969 [[arXiv:1912.04768](#) [hep-th]].
9. **“Aspects of nonsupersymmetric string phenomenology,”** J. Rizos, *Int. J. Mod. Phys. A* **33** (2018) no.34, 1845002 doi:10.1142/S0217751X18450021.
10. **“Classification of left–right symmetric heterotic string vacua,”** A. E. Faraggi, G. Harries and J. Rizos, *Nucl. Phys. B* **936** (2018) 472 doi:10.1016/j.nuclphysb.2018.09.028 [[arXiv:1806.04434](#) [hep-th]].
11. **“Classification of Standard-like Heterotic-String Vacua,”** A. E. Faraggi, J. Rizos and H. Sonmez, *Nucl. Phys. B* **927** (2018) 1 doi:10.1016/j.nuclphysb.2017.12.006 [[arXiv:1709.08229](#) [hep-th]].
12. **“Wilsonian dark matter in string derived Z' model,”** L. Delle Rose, A. E. Faraggi, C. Marzo and J. Rizos, *Phys. Rev. D* **96** (2017) no.5, 055025 doi:10.1103/PhysRevD.96.055025 [[arXiv:1704.02579](#) [hep-ph]].
13. **“A Solution to the Decompactification Problem in Chiral Heterotic Strings,”** I. Florakis and J. Rizos, *Nucl. Phys. B* **921** (2017) 1 doi:10.1016/j.nuclphysb.2017.05.002 [[arXiv:1703.09272](#) [hep-th]].
14. **“Chiral Heterotic Strings with Positive Cosmological Constant,”** I. Florakis and J. Rizos, *Nucl. Phys. B* **913** (2016) 495 doi:10.1016/j.nuclphysb.2016.09.018 [[arXiv:1608.04582](#) [hep-th]].
15. **“The 750 GeV di-photon LHC excess and extra Z 's in heterotic-string derived models”** A. E. Faraggi and J. Rizos, *Eur. Phys. J. C* **76** (2016) no.3, 170 [[arXiv:1601.03604](#) [hep-ph]].
16. **“A light Z' heterotic-string derived model,”** A. E. Faraggi and J. Rizos, *Nucl. Phys. B* **895** (2015) 233 [[arXiv:1412.6432](#) [hep-th]].
17. **“Proton Stability in $SU(5) \times U(1)$ and $SU(6) \times SU(2)$ GUTs”** A. E. Faraggi, M. Paraskevas, J. Rizos and K. Tamvakis, *Phys. Rev. D* **90** (2014) 015036 [[arXiv:1405.2274](#) [hep-ph]].
18. **“Genetic Algorithms and the Search for Viable String Vacua”** S. Abel and J. Rizos, *JHEP* **1408** (2014) 010 [[arXiv:1404.7359](#) [hep-th]].
19. **“Classification of Flipped $SU(5)$ Heterotic-String Vacua,”** A. E. Faraggi, J. Rizos and H. Sonmez, *Nucl. Phys. B* **886** (2014) 202–242 [[arXiv:1403.4107](#) [hep-th]].
20. **“Top quark mass coupling and classification of weakly-coupled heterotic superstring vacua”** J. Rizos, *Eur. Phys. J. C* **74**, 2905 (2014) [[arXiv:1404.0819](#) [hep-ph]].
21. **“Shock waves as branes with throats”** J. Rizos and N. Tetradis, *JHEP* **1302** (2013) 112 [[arXiv:1210.4730](#) [hep-th]].
22. **“String Derived Exophobic $SU(6) \times SU(2)$ GUTs”** L. Bernard, A. E. Faraggi, I. Glasser, J. Rizos and H. Sonmez, *Nucl. Phys. B* **868** (2013) 1 [[arXiv:1208.2145](#) [hep-th]].
23. **“Classicalization as a tunnelling phenomenon,”** J. Rizos, N. Tetradis and G. Tsolias, *JHEP* **1208** (2012) 054 [[arXiv:1206.3785](#) [hep-th]].
24. **“Dynamical Classicalization,”** J. Rizos and N. Tetradis, *JHEP* **1204** (2012) 110, [[arXiv:1112.5546](#) [hep-th]].
25. **“On the dynamics of classicalization,”** N. Brouzakis, J. Rizos and N. Tetradis, *Phys. Lett. B* **708** (2012) 170–173, [[arXiv:1109.6174](#) [hep-th]].
26. **“Top Quark Mass in Exophobic Pati–Salam Heterotic String Model,”** Kyriakos Christodoulides, Alon E. Faraggi,

- John Rizos, *Phys. Lett.* **B702** (2011) 81-89 [[arXiv: hep-ph/1104.2264](#)]
27. “**Classification of Heterotic Pati–Salam Models**”, B. Assel, K. Christodoulides, A. E. Faraggi, C. Kounnas and J. Rizos, *Nucl. Phys.* **B844** (2011) 365-396 [[arXiv: hep-th/1007.2268](#)].
 28. “**Hierarchical Neutrino Masses and Mixing in Flipped-SU(5)**”, J. Rizos and K. Tamvakis, *Phys. Lett.* **B685** (2010) 67-71 [[arXiv: hep-th/0912.3997](#)].
 29. “**Exophobic Quasi-Realistic Heterotic String Vacua**”, B. Assel, K. Christodoulides, A. E. Faraggi, C. Kounnas and J. Rizos, *Phys. Lett.* **B683** (2010) 306 [[arXiv: hep-th/0910.3697](#)].
 30. “**Spinor-Vector Duality in Heterotic SUSY Vacua**”, Tristan Catelin-Jullien, A. E. Faraggi, C. Kounnas and J. Rizos, *Nucl. Phys.* **B812** (2009) 103-127 [[arXiv: hep-th/0807.4084](#)].
 31. “**Spinor-vector duality in N=2 heterotic string vacua**”, A. E. Faraggi, C. Kounnas and J. Rizos, *Nucl. Phys.* **B799** (2008) 19-33 [[arXiv: hep-th/07120747](#)].
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